Foundations of Logic and Proofs

CS-123: Quantitative Reasoning–I

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The Foundations: Logic and Proofs

- The rules of logic give **precise meaning** to mathematical statements.
- These rules let us distinguish between **valid** and **invalid** arguments.
- Logic is the basis of **all** mathematical reasoning – and of all **automated** reasoning.

Why This Matters

Logic has direct applications to: designing computer circuits, specifying systems, artificial intelligence, computer programming, programming languages, and many other fields.

Definition: Proposition

Definition

A **proposition** is a declarative sentence (a sentence that declares a fact) that is either **true** or **false**, but not both.

- Propositions are the basic building blocks of logic.
- We call the area of logic dealing with propositions **propositional logic** (or the *propositional calculus*).

Example 1: These ARE Propositions

Example 1

All of the following declarative sentences are propositions.

1. Washington, D.C., is the capital of the United States of America.
2. Toronto is the capital of Canada.
3. $1 + 1 = 2$.
4. $2 + 2 = 3$.

Solution: Propositions 1 and 3 are **true**; propositions 2 and 4 are **false**. Every one of these sentences has a definite truth value, so all four are propositions.

Example 2: These Are NOT Propositions

Example 2

Consider: (1) What time is it? (2) Read this carefully. (3) $x + 1 = 2$. (4) $x + y = z$.

Solution: Sentences 1–2 are **not** propositions – not declarative (a question and a command). Sentences 3–4 are **not** propositions – neither true nor false until we assign values.

Propositional Variables and Truth Values

- We use letters to denote **propositional variables** (also called **sentential variables**): the conventional letters are $p, q, r, s, \ldots$
- The **truth value** of a proposition is **true**, denoted T, if it is a true proposition, and **false**, denoted F, if it is a false proposition.
- A proposition that **cannot** be broken down into simpler propositions is called an **atomic proposition**.

Compound Propositions

New propositions, called **compound propositions**, are formed from existing propositions using **logical operators** (also called **connectives**).

Definition 2: Atomic Statements

Definition 2 (Atomic Statements)

Declarative sentences which **cannot** be further split into simpler sentences are called **atomic statements** (also called **primary statements** or **primitive statements**).

Example

"p is a prime number."

- An atomic statement makes just **one** claim – there is nothing smaller inside it to break apart with a connective.
- These are the "raw ingredients" we combine using $\neg, \wedge, \vee, \oplus, \rightarrow, \leftrightarrow$.

Definition 3: Compound Statements

Definition 3 (Compound Statements)

New statements can be formed from atomic statements through the use of **connectives** such as "and," "but," "or," etc. The resulting statement is called a **molecular** or **compound** (composite) statement.

Example

"If p is a prime number, then the divisors of p are p and 1 itself."

Connecting the Dots

This is exactly $p \rightarrow q$ where p: "p is a prime number," q: "the divisors of p are p and 1 itself" – two atomic statements joined by a conditional connective.

Practice: Which Are Propositions?

✓ Question

Determine which are propositions, and give the truth value of each proposition.

1. Boston is the capital of Massachusetts.
2. Miami is the capital of Florida.
3. $2 + 3 = 5$.
4. $5 + 7 = 10$.

Answers:

1. Proposition, **True** (Boston is the capital of MA).
2. Proposition, **False** (Tallahassee is the capital of FL).
3. Proposition, **True**.
4. Proposition, **False** ($5 + 7 = 12$).

Practice: Which Are Propositions? (Try It)

✎ Unsolved – Try These

Which of these are propositions? For those that are, give the truth value.

1. Do not pass go.
2. There are no black flies in Maine.
3. $4 + x = 5$.
4. The moon is made of green cheese.

Practice: Which Are Propositions? (Try It)

✎ Unsolved – Try These

Which of these are propositions? For those that are, give the truth value.

1. Do not pass go.
2. There are no black flies in Maine.
3. $4 + x = 5$.
4. The moon is made of green cheese.

Answers: 1) Not a proposition (a command). 2) Proposition (false – Maine does have black flies). 3) Not a proposition (depends on x). 4) Proposition (false).

Can a Sentence Break Logic Itself?

The Liar Paradox

Consider: **"This statement is false."**

- If it's **true**, then what it says is correct – so it *is* false. Contradiction!
- If it's **false**, then what it says is incorrect – so it *is* true. Contradiction!

The Resolution

This sentence **cannot** be assigned a consistent truth value – so it is **not a proposition** at all, by our definition. This shows why the definition of "proposition" must be taken carefully.

The Barber Paradox

An Old Puzzle

A town's only barber shaves **everyone who does not shave themselves, and no one else**. Does the barber shave himself?

- If he shaves himself, then (by the rule) he must be someone who does *not* shave himself – contradiction.
- If he does *not* shave himself, then (by the rule) the barber *must* shave him – contradiction.

The Barber Paradox

An Old Puzzle

A town's only barber shaves **everyone who does not shave themselves, and no one else**. Does the barber shave himself?

- If he shaves himself, then (by the rule) he must be someone who does *not* shave himself – contradiction.
- If he does *not* shave himself, then (by the rule) the barber *must* shave him – contradiction.

Conclusion: No such barber can exist – the description is self-contradictory, just like the Liar Paradox.

Connective 1: Negation (¬)

Definition 2 – Negation

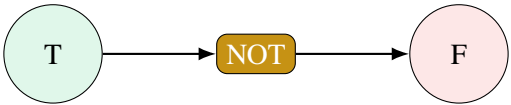
The negation of p, written $\neg p$, is the statement "It is not the case that p." Its truth value is the **opposite** of p.

p	$\neg p$
T	F
F	T

Everyday example

p: "Ali's laptop has 16 GB of RAM."

$\neg p$: "Ali's laptop does *not* have 16 GB of RAM."



Connective 2: Conjunction ($\wedge$) – "AND"

Definition 3 – Conjunction

The conjunction of p and q, written $p \wedge q$, is the proposition "p and q." It is **true only when both** p and q are true.

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

p: "Sara's phone has a good camera."
q: "Sara's phone has long battery life."
$p \wedge q$: "Sara's phone has a good camera *and* long battery life."

Note: the word "but" also signals AND – "It's sunny, but it's cold" means sunny $\wedge$ cold.

Connective 3: Disjunction ($\vee$) – "OR"

Definition 4 – Disjunction

The disjunction of p and q, written $p \vee q$, is the proposition "p or q." It is **false only when both** p and q are false. (This is the *inclusive* or.)

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

p: "You have a NUST student card."
q: "You have a NUST staff card."
$p \vee q$: You are allowed into the library if *either* (or both) hold.

Connective 4: Exclusive OR ($\oplus$)

Definition 5 – Exclusive Or

$p \oplus q$ is true when **exactly one** of p, q is true (not both).

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

Menu example: "Soup *or* salad comes with the entrée" – you may pick one, not both. This is $\oplus$, not $\vee$.

Spot the difference

A restaurant coupon says: *"Free drink with any burger or fries."* Is this $\vee$ or $\oplus$? Would you get a free drink if you order *both*?

Examples – Evaluating Connectives

✓ Solved

Let p: "$5 > 3$" (T) and q: "$2 + 2 = 5$" (F). Evaluate each:

- $\neg p$: negate T $\Rightarrow$ **F**
- $p \wedge q$: T and F $\Rightarrow$ **F**
- $p \vee q$: T or F $\Rightarrow$ **T**
- $p \oplus q$: exactly one true (only p) $\Rightarrow$ **T**
- $\neg p \wedge \neg q$: F and T $\Rightarrow$ **F**

Practice – Evaluating Connectives



Let p: "Islamabad is the capital of Pakistan" (T) and q: "10 is a prime number" (F). Evaluate:

1. $p \vee q$
2. $\neg q$
3. $p \wedge \neg q$
4. $p \oplus q$
5. $\neg p \vee \neg q$

Practice – Evaluating Connectives



Let p: "Islamabad is the capital of Pakistan" (T) and q: "10 is a prime number" (F). Evaluate:

1. $p \vee q$
2. $\neg q$
3. $p \wedge \neg q$
4. $p \oplus q$
5. $\neg p \vee \neg q$

Answers: 1) T 2) T 3) T 4) T 5) T

Visual Summary: The Four Connectives So Far



NOT AND (inclusive) OR exclusive OR

Same building blocks $\longrightarrow$ your CPU's logic gates!

(AND, OR, NOT gates in digital circuits are literally these connectives.)

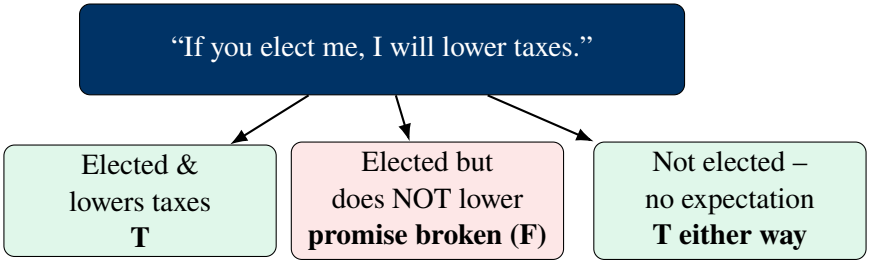
Connective 5: The Conditional ($\rightarrow$)

Definition 6 – Conditional Statement

Let p, q be propositions. The conditional statement $p \rightarrow q$ ("if p then q") is **false only when** p is true and q is false; otherwise it is true.
p is called the **hypothesis** (antecedent), q the **conclusion** (consequent).

p	q	$p \rightarrow q$
T	T	T
T	F	**F**
F	T	T
F	F	T

Intuition: The "Broken Promise" Picture



- Think of $p \to q$ as a **contract/promise**: it is broken *only* when the condition p holds but the promised outcome q fails.
- If p never happens, you can't call the promise "broken" – vacuously true.

Many Faces of "$p \rightarrow q$" in English

- if p, then q
- if p, q
- p implies q
- p is sufficient for q
- p only if q
- q if p
- q when p
- q whenever p
- a sufficient condition for q is p
- a necessary condition for p is q
- q is necessary for p
- q follows from p
- q unless $\neg p$
- q provided that p

Quick Poll

"You can get an A only if you score at least 90% on the final."
Let p : "you get an A", q : "you score $\geq 90\%$". Is this $p \rightarrow q$ or $q \rightarrow p$?

Examples – Translating $p \to q$

✓ Solved

Let p: "Bilal studies discrete mathematics" and q: "Bilal gets a good job."
Express $p \to q$ in English (several ways):

- "If Bilal studies discrete mathematics, then he gets a good job."
- "Bilal gets a good job whenever he studies discrete mathematics."
- "For Bilal to get a good job, it is sufficient that he studies discrete mathematics."
- "Bilal gets a good job unless he does not study discrete mathematics."

✓ Solved – Evaluating a Conditional

Let p: "$3 > 5$" (F), q: "8 is even" (T). Find $p \to q$.
Solution: p is F, so regardless of q, $p \to q$ is **T** (vacuously true).

Practice – Conditional Statements



Let p: "you charge your phone" and q: "your phone turns on."

1. Write $p \rightarrow q$ using *three* different English phrasings.
2. Let p: "4 is odd" (F), q: "9 is odd" (T). Evaluate $p \rightarrow q$.
3. Let p: "4 is even" (T), q: "$4 > 10$" (F). Evaluate $p \rightarrow q$.
4. "A necessary condition for Sara to graduate is that she pays her dues." Write this as $p \rightarrow q$, clearly stating p and q.

Practice – Conditional Statements

Let p: "you charge your phone" and q: "your phone turns on."

1. Write $p \to q$ using *three* different English phrasings.
2. Let p: "4 is odd" (F), q: "9 is odd" (T). Evaluate $p \to q$.
3. Let p: "4 is even" (T), q: "$4 > 10$" (F). Evaluate $p \to q$.
4. "A necessary condition for Sara to graduate is that she pays her dues." Write this as $p \to q$, clearly stating p and q.

Answers: 2) T (vacuous, p false) 3) F ($T \to F$ is the only false case) 4) p: Sara graduates, q: Sara pays her dues; statement is $p \to q$.

A Curious Case: Vacuous Truth

Careful – Logic ≠ everyday cause-and-effect

"If Ahmed has a smartphone, then $2 + 3 = 5$." is **TRUE** – because the conclusion is true, the whole statement is true regardless of the hypothesis or any real connection.

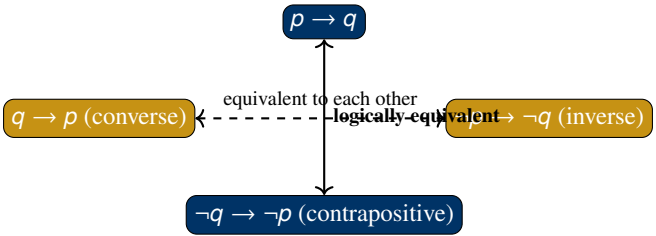
"If Ahmed has a smartphone, then $2 + 3 = 6$." is **TRUE if Ahmed has no smartphone** – even though $2 + 3 = 6$ is false!

- Mathematical logic does **not** require a cause-and-effect relationship between p and q – only the truth-value rule in the table matters.
- This differs from `if` in programming: `if (p) then S` *executes* a block S; here $p \to q$ is itself just a statement with a truth value.

Converse, Inverse, Contrapositive

Starting from $p \to q$, we build three related statements:

Name	Form	Same truth value as $p \to q$?
Converse	$q \to p$	No
Inverse	$\neg p \to \neg q$	No
Contrapositive	$\neg q \to \neg p$	**Yes, always**



Example – Converse / Inverse / Contrapositive

✓ Solved

Statement: "If a number is divisible by 4, then it is divisible by 2."
p: number divisible by 4; q: number divisible by 2.

- **Converse** ($q \to p$): "If a number is divisible by 2, then it is divisible by 4." (False, e.g. 6)
- **Inverse** ($\neg p \to \neg q$): "If a number is not divisible by 4, then it is not divisible by 2." (False, e.g. 6)
- **Contrapositive** ($\neg q \to \neg p$): "If a number is not divisible by 2, then it is not divisible by 4." (**True, always**)

Practice: Converse / Inverse / Contrapositive

Group Activity – 3 minutes

Statement: "If it snows in Islamabad, then classes are cancelled."

1. Write the **converse**.
2. Write the **inverse**.
3. Write the **contrapositive**.
4. Which of these must have the *same* truth value as the original? Why?

Practice: Converse / Inverse / Contrapositive

Group Activity – 3 minutes

Statement: "If it snows in Islamabad, then classes are cancelled."

1. Write the **converse**.
2. Write the **inverse**.
3. Write the **contrapositive**.
4. Which of these must have the *same* truth value as the original? Why?

Compare answers with the pair next to you, then we discuss as a class.

Connective 6: The Biconditional ($\leftrightarrow$)

Definition 7 – Biconditional

The biconditional $p \leftrightarrow q$ ("p if and only if q", abbreviated **iff**) is **true exactly when p and q have the same truth value**.

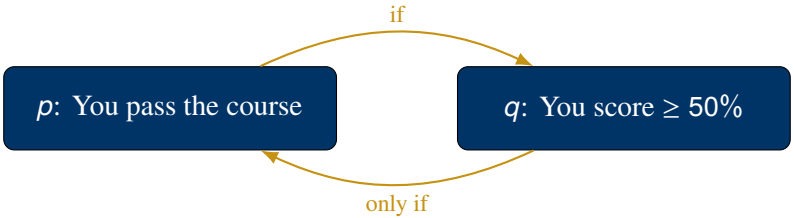
p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Key insight: $p \leftrightarrow q$ is the same as

$$(p \rightarrow q) \wedge (q \rightarrow p)$$

– *both directions* must hold. That's why we read it as "p iff q."

Biconditional – Everyday Reading



English phrasings of $p \leftrightarrow q$:

- "p if and only if q"
- "p is necessary and sufficient for q"
- "if p then q, and conversely"

Quick Poll

"A number is even if and only if it is divisible by 2."

Is there any case where one side is true and the other false? Try $n = 7$ and $n = 8$.

Examples – Biconditional

✓ Solved

Q1. Let p: "n is even" and q: "n is divisible by 2." Is $p \leftrightarrow q$ always true?
A. Yes – for every integer n, "even" and "divisible by 2" always agree $\Rightarrow p \leftrightarrow q$ is a **tautology** (always T).

Q2. Let p: "triangle ABC is equilateral" (F for a given triangle) and q: "triangle ABC has three equal angles" (F for the same triangle). Evaluate $p \leftrightarrow q$.
A. Both F $\Rightarrow$ same truth value $\Rightarrow p \leftrightarrow q = \mathbf{T}$.

Practice – Biconditional

✎ Try It Yourself

1. Let p: "15 is prime" (F), q: "15 is odd" (T). Evaluate $p \leftrightarrow q$.
2. Let p: "today is Friday," q: "tomorrow is Saturday." Is $p \leftrightarrow q$ always true? Explain in one line.
3. Write $p \leftrightarrow q$ in words for p: "you have a valid ID," q: "you may enter the exam hall."

Answers: 1) F (T, F differ) 2) Yes, always true – the two events are logically tied together 3) "You may enter the exam hall if and only if you have a valid ID."

Constructing Truth Tables: The Method

How to Build a Truth Table

1. Count the propositional variables ($p, q, r, \ldots$). With n variables, the table needs $\mathbf{2^n}$ **rows** (every possible combination of T/F).
2. List all combinations in the standard order (alternate T/F fastest in the rightmost variable).
3. Add one column at a time for each **piece** of the expression, working from the innermost brackets outward, in this order: $\neg \rightarrow \wedge \rightarrow \vee \rightarrow \rightarrow \rightarrow \leftrightarrow$.
4. The **last column** you build is the truth table for the whole expression.

✎ Quick Check

How many rows does a truth table need for **4** variables?
Answer: $2^4 = 16$ rows.

Example: 2-Variable Truth Table

Construct the truth table for $\neg p \vee q$.

p	q	$\neg p$	$\neg p \vee q$
T	T	F	T
T	F	F	F
F	T	T	T
F	F	T	T

Example: 3-Variable Truth Table

Construct the truth table for $(p \wedge q) \vee \neg r$.

p	q	r	$p \wedge q$	$\neg r$	$(p \wedge q) \vee \neg r$
T	T	T	T	F	T
T	T	F	T	T	T
T	F	T	F	F	F
T	F	F	F	T	T
F	T	T	F	F	F
F	T	F	F	T	T
F	F	T	F	F	F
F	F	F	F	T	T

3 variables $\Rightarrow 2^3 = 8$ rows. Build $p \wedge q$ first, then $\neg r$, then combine the two middle columns with $\vee$ to get the final column.

Practice: Construct These Truth Tables

✎ Try It Yourself

Build the full truth table (showing all intermediate columns) for each expression below.

1. $p \wedge \neg q$ (2 variables, 4 rows)
2. $p \rightarrow (q \vee r)$ (3 variables, 8 rows)
3. $(p \leftrightarrow q) \wedge \neg p$ (2 variables, 4 rows)

Final-column answers to check your work:
1) T, F, F, F (rows TT, TF, FT, FF)
2) T,T,T,T,T,F,T,F (rows TTT,TTF,TFT,TFF,FTT,FTF,FFT,FFF)
3) F, F, F, T (rows TT, TF, FT, FF)

Putting It All Together

Connective	**Read as**	**Symbol**
Negation	not p	$\neg p$
Conjunction	p and q	$p \wedge q$
Disjunction	p or q	$p \vee q$
Exclusive or	p xor q	$p \oplus q$
Conditional	if p then q	$p \rightarrow q$
Biconditional	p iff q	$p \leftrightarrow q$

Let p: "it rains," q: "the ground is wet," with $p = T, q = T$ today. Evaluate:

1. $p \rightarrow q$
2. $q \rightarrow p$ (the converse)
3. $\neg p \vee q$
4. $p \leftrightarrow q$

Looking Ahead – Week 2

- Translating English sentences $\longrightarrow$ symbolic logic
- System specifications & logic puzzles
- Logic circuits (AND/OR/NOT gates $\leftrightarrow$ connectives)
- Propositional equivalences using truth tables, De Morgan's Laws
- Tautologies and contradictions

Homework: Read Rosen, Section 1.1, and attempt Exercises 1–15 before next class.

Bits and Boolean Variables

Definitions

A **bit** is a symbol with two possible values: 0 (zero) or 1 (one). "Bit" comes from *binary digit*. A variable is called a **Boolean variable** if its value is either true or false.

Truth Value	Bit
T	1
F	0

A bit can represent a truth value: **1 represents T, 0 represents F** – this is the bridge between logic and computer hardware.

Bit Operators: OR, AND, XOR

From Logic to Bits

Replace T with 1 and F with 0 in the truth tables for $\vee, \wedge, \oplus$ to get the corresponding **bit operations**. Programming languages call these `OR`, `AND`, `XOR`.

x	y	$x \vee y$	$x \wedge y$	$x \oplus y$
0	0	0	0	0
0	1	1	0	1
1	0	1	0	1
1	1	1	1	0

Bit Strings

Definition 7

A **bit string** is a sequence of zero or more bits. The **length** of the string is the number of bits in it.

Example 15

101010011 is a bit string of length **9**.

Bitwise Operations

Given two bit strings of the *same* length, the **bitwise OR**, **bitwise AND**, and **bitwise XOR** apply the corresponding bit operation to each pair of matching bits.

Example 16: Bitwise Operations

Example 16

Find the bitwise OR, AND, and XOR of 0110110110 and 1100011101.

01 1011 0110	
11 0001 1101	
11 1011 1111	(bitwise OR)
01 0001 0100	(bitwise AND)
10 1010 1011	(bitwise XOR)

Each result bit is the OR/AND/XOR of the two bits directly above it.

Practice: Bitwise Operations

✎

Find the bitwise OR, AND, and XOR of 1011110 and 0100001.

✎ **Unsolved – Try This**

Evaluate: $(0101 \wedge 1101) \vee 0100$.

Practice: Bitwise Operations

✎

Find the bitwise OR, AND, and XOR of 1011110 and 0100001.
OR: 1111111
AND: 0000000
XOR: 1111111

✎ **Unsolved – Try This**

Evaluate: $(0101 \wedge 1101) \vee 0100$.
Step 1: $0101 \wedge 1101 = 0101$. **Step 2:** $0101 \vee 0100 = 0101$.

Practice: Longer Bit Strings

✎ Unsolved – Try This

Find the bitwise OR, AND, and XOR of
11 1111 1111 and 00 0000 0000.

Practice: Longer Bit Strings

✎ Unsolved – Try This

Find the bitwise OR, AND, and XOR of
11 1111 1111 and 00 0000 0000.
OR: 11 1111 1111
AND: 00 0000 0000
XOR: 11 1111 1111

Notice: OR-ing with all zeros changes nothing; AND-ing with all zeros wipes everything to 0 – exactly like $p \vee F \equiv p$ and $p \wedge F \equiv F$ in logic!